

CO-2 - Assignment-2.

- 1) DES Problem: DES (Data Encryption Standard) is a symmetric block cipher that operates on a 64 bit plain text using a 56-bit effective key. DES uses a Feistel structure with 16 rounds.

STEP 1: Initial permutation

64 bits convert into two 32-bits

$$I.P(P_T) = L_0 \parallel R_0$$

$L_0 \Rightarrow$ Left 32 bits, $R_0 \Rightarrow$ Right 32 bits.

STEP 2: Expansion of R_0

32-bit $\longrightarrow$ 48 bits

R_0 (32 bits) $\longrightarrow$ $E(R_0)$ (48 bits)

STEP 3: XOR with round key.

$$X = E(R_0) \oplus K_1$$

STEP 4: S-box substitution.

48 bits $\longrightarrow$ 62 bits

Passed through a DES S-box.

48 bits $\longrightarrow$ 32 bits

STEP 5: P-box permutation

32 bit are rearranged using the

P-permutation.

$$F(R_0, k_1) = P(S(E(R_0) \oplus k_1))$$

STEP 6 : XOR with L_0

$$R_1 = L_0 \oplus F(R_0, k_1)$$

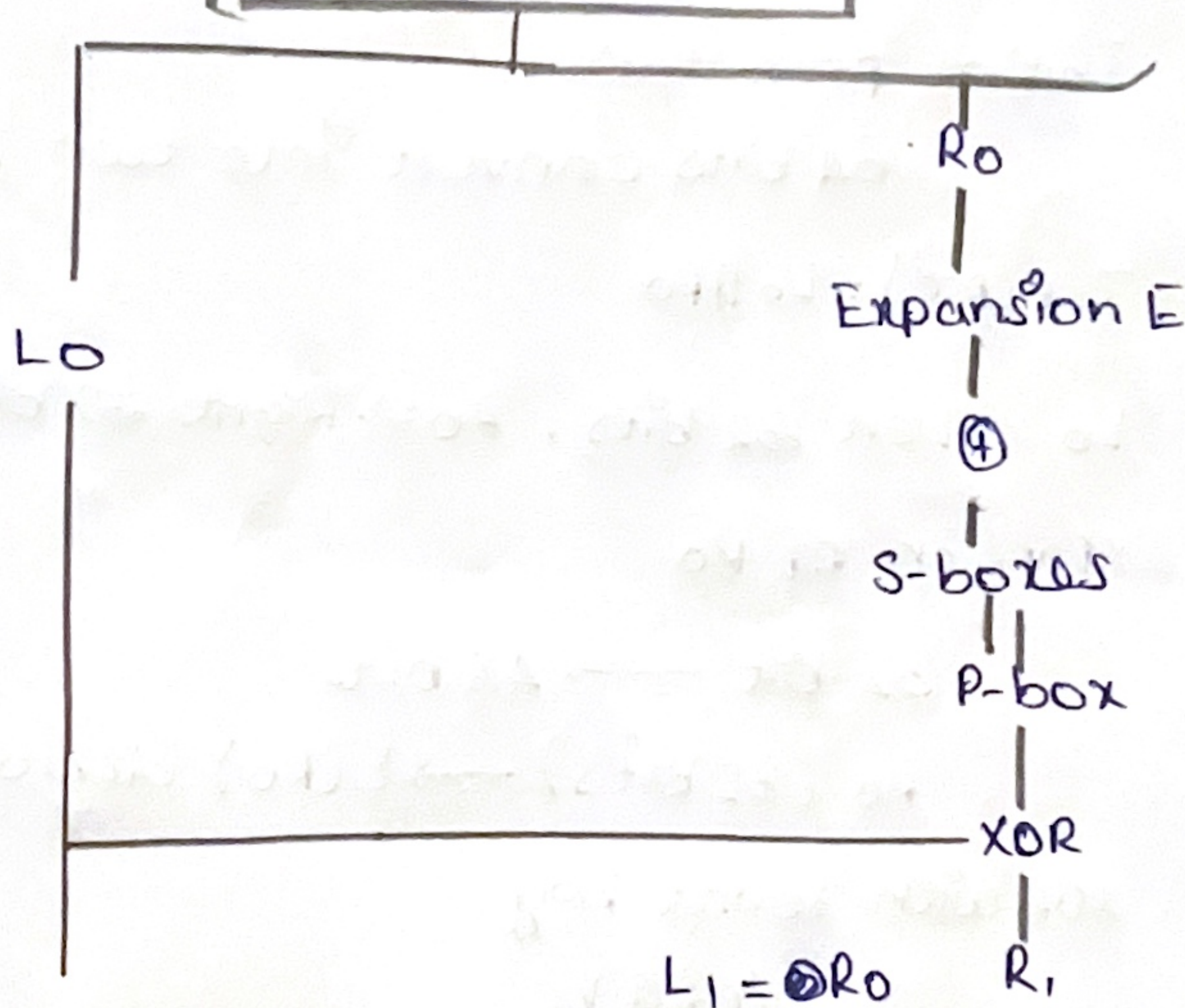
$$L_1 = R_0$$

one DES round:

$$R_1 = L_0 \oplus F(R_0, k_1)$$

Plain text

Initial permutation



2) compare symmetric and Asymmetric Encryption using real time application:

=> Encryption techniques are mainly classified into symmetric encryption and asymmetric encryption.

STEP 3: Select Public Exponent:

$$e = 7$$

$$\gcd(7, 60) = 1$$

$$\boxed{\therefore e = 7}$$

calculate Private key:

$$ed = 1 \pmod{60}$$

$$7d \equiv 1 \pmod{60}$$

$$7 \times 43 = 301$$

$$301 \pmod{60} = 1$$

RSA keys:

Public key

Private key

$$(e, n) = (7, 77)$$

$$(d, n) = (43, 77)$$

RSA Encryption:

$$C = M^e \pmod{n} \Rightarrow C = 9^7 \pmod{77}$$

$$9^2 = 81 \equiv 4 \pmod{77}$$

$$576 \pmod{77} = 37$$

$$\boxed{\therefore C = 37}$$

$\therefore$ Encrypted ciphertext = 37.

4) A CBC Mod c: (cipher block chaining)

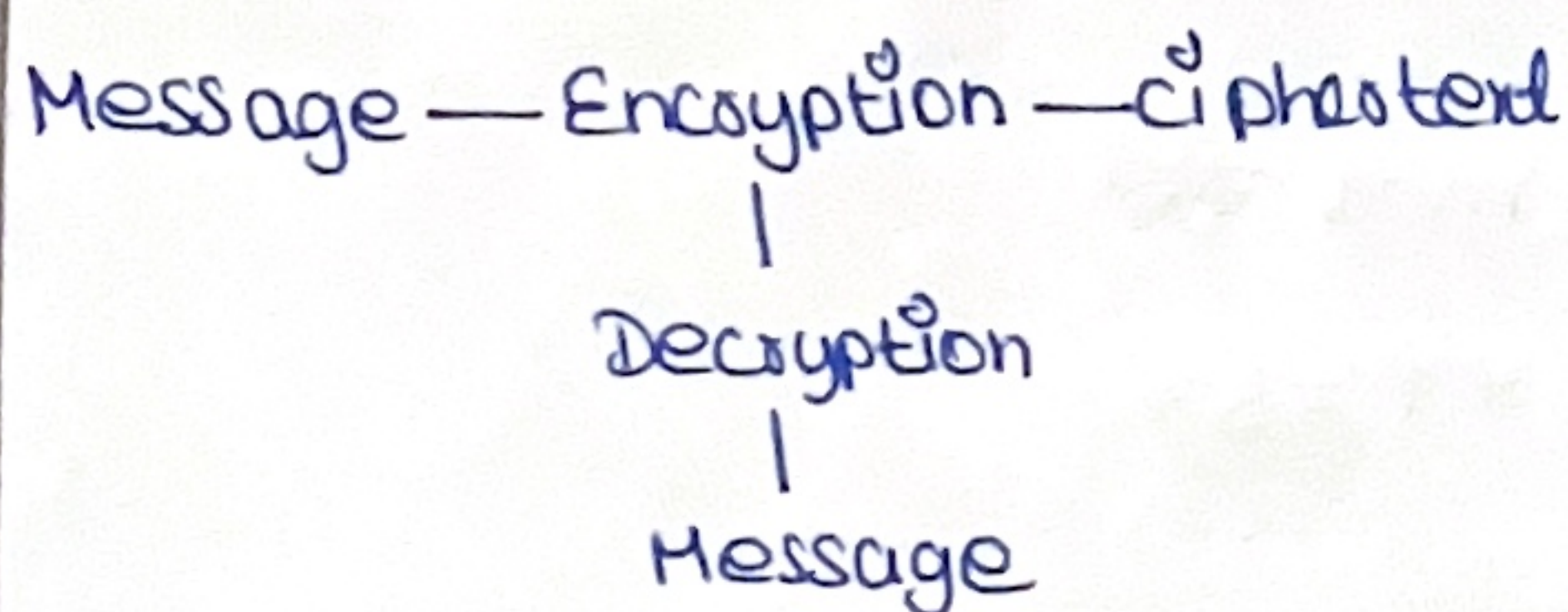
In CBC each plaintext block is XOR with the previous cipher block before Encryption.

1st block $\rightarrow$ (IV) is used.

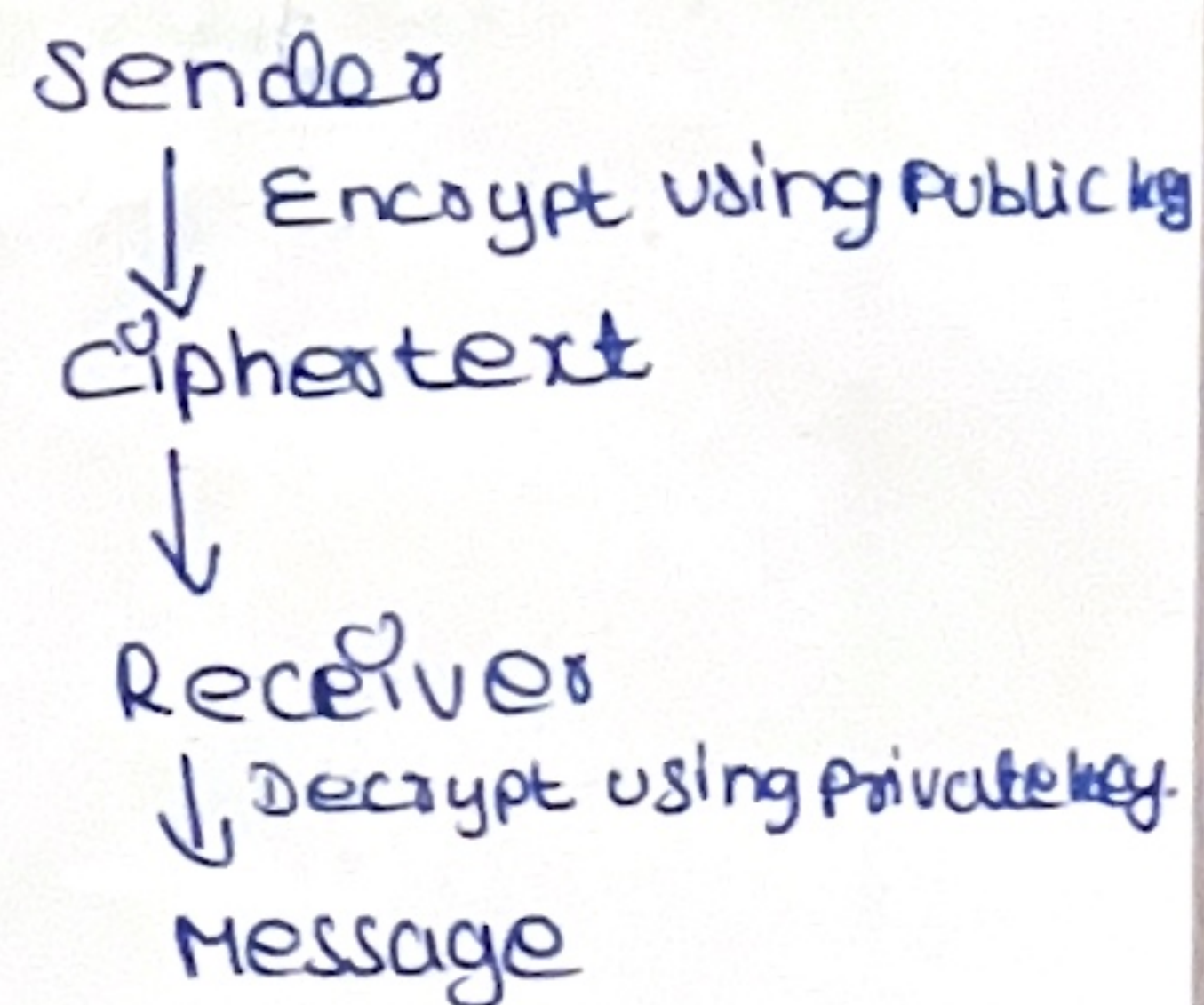
Feature	Symmetric Encryption	Asymmetric Encryption.
key used	one shared key	Public and private key
Encryption/Decryption	same key	Different key
Speed	very fast	comparatively slow
scalable	Large amount of data	key exchange and authentication.
Examples	AES, DES, ChaCha20	RSA, ECC

Symmetric Encryption

Secret key



Asymmetric Encryption



3) Given:

$$P = 7, q = 11, M = 9$$

$$n = P \times q$$

$$7 \times 11 = 77$$

$$\phi(n) = (P-1)(q-1)$$

$$= (7-1)(11-1) = 6 \times 10$$

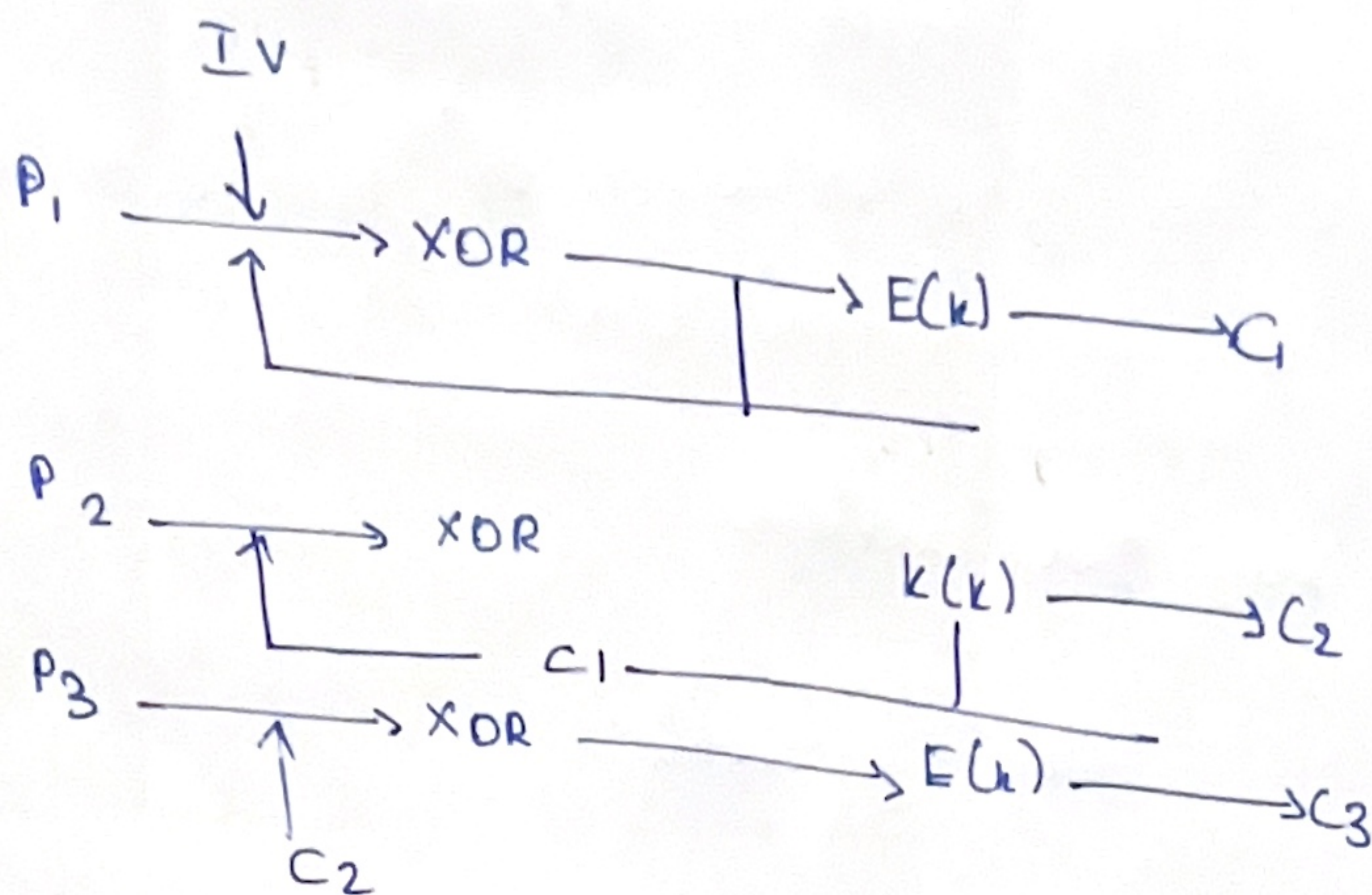
$$\phi(n) = 60$$

Encryption:-

$$C_i = E_k [P_i \oplus C_{i-1}]$$

$$\therefore C_0 = IV$$

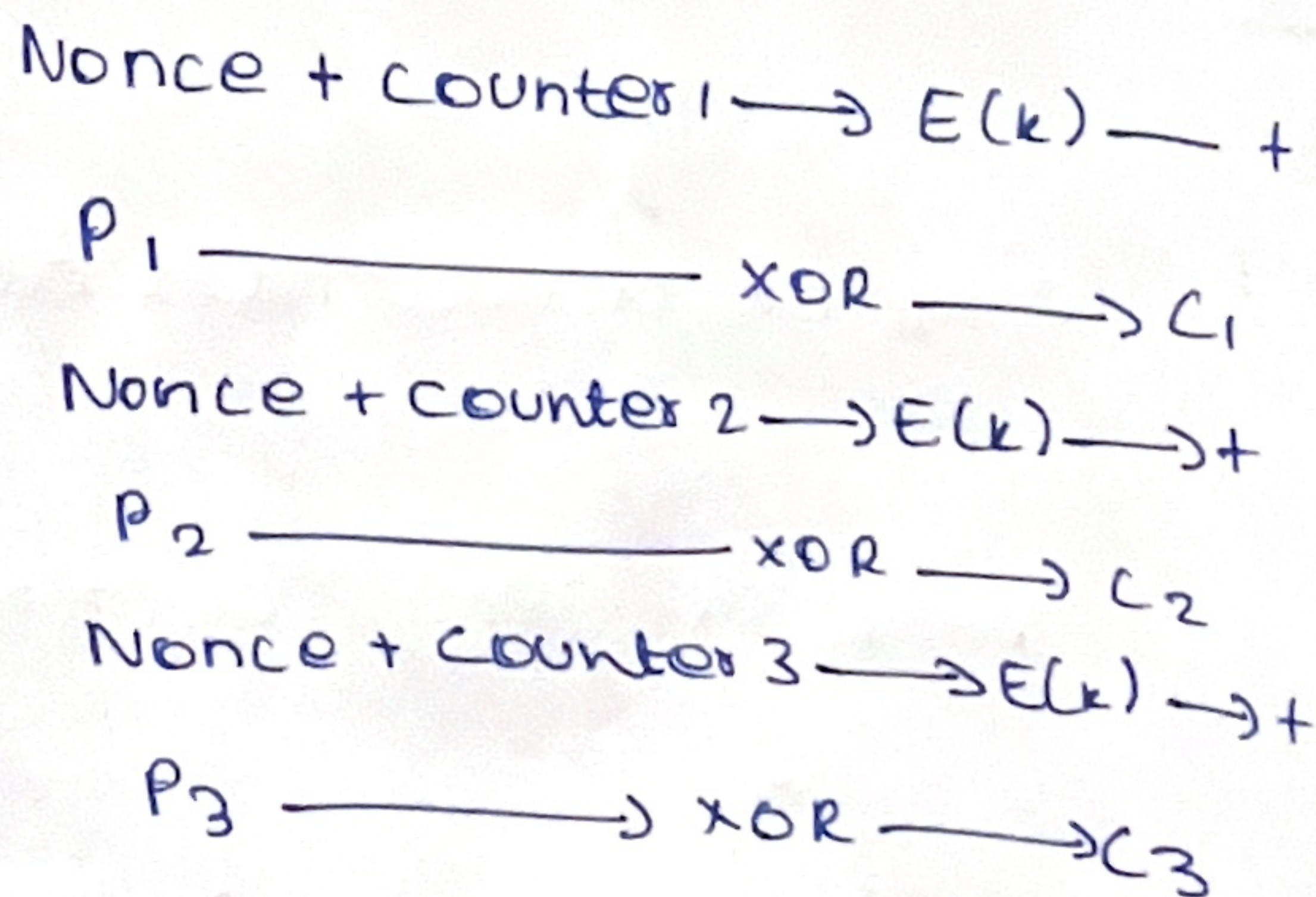
CBC:



B. CTR Mode: (Counter mode)

$$C_i = P_i \oplus E_k (\text{Nonce} \parallel \text{counter})$$

CTR Encryption Diagram:



CTR Mode:

CTR is more suitable for parallel processing because each counter block can be encrypted independently.

5) Quantum Cryptography:-

⇒ Quantum Cryptography is a security technique that uses the principle of Quantum mechanics to provide secure communication.

Quantum key Distribution:-

